

South African Fraud Detection Using Machine Learning: Summary Report

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Project Type: Data Science / Machine Learning

Dataset: Synthetic (adapted to South African transaction behaviour)

1. Introduction

Fraudulent transactions represent a critical challenge for financial institutions and digital commerce platforms in South Africa. As transaction volumes increase and payment systems become more digitised, traditional rule-based fraud detection systems struggle to adapt to evolving fraud patterns. Machine learning offers a scalable and adaptive approach to identifying suspicious activity while balancing customer experience and regulatory requirements.

This project applies machine learning techniques to develop a fraud detection model using a synthetic dataset designed to reflect realistic South African transaction behaviour. The goal is not only to achieve strong predictive performance, but also to ensure explainability and business relevance, which are essential in regulated environments such as banking and financial services.

2. Business Understanding

The core business problem addressed in this project is the early and accurate identification of fraudulent transactions. Fraud events are rare, but their financial and reputational impact is significant. A successful fraud detection system must therefore prioritise identifying fraudulent activity while controlling the number of legitimate transactions incorrectly flagged as fraud.

The key objectives of the project are:

- To maximise fraud detection (high recall)
- To maintain acceptable precision to avoid excessive false alerts
- To provide transparent model explanations to support trust and compliance

- To simulate a realistic South African fraud detection scenario

3. Data Understanding

The dataset used in this project is synthetic, as real fraud data is typically unavailable due to privacy laws such as POPIA and institutional confidentiality. However, the dataset was adapted to reflect South African market characteristics, including transaction values, behavioural patterns, and realistic fraud ratios.

The target variable, **is_fraud**, is binary:

- **1** indicates a fraudulent transaction
- **0** indicates a legitimate transaction

Initial exploration revealed:

- Severe class imbalance, with fraud representing a small minority of transactions
- A mixture of numerical and categorical variables
- Time-based and amount-based features with potential predictive value

4. Data Wrangling and Exploratory Data Analysis (EDA)

Data cleaning steps included validating data types, checking for missing values, and ensuring consistency across features. Exploratory data analysis was then conducted to understand transaction behaviour and fraud patterns.

Key **EDA** insights include:

- **Class imbalance:** Fraud cases are significantly underrepresented, confirming the need for imbalance-handling techniques.
- **Transaction amounts:** Fraudulent transactions tend to cluster within certain ranges, while legitimate transactions show broader variability.
- **Time patterns:** Fraud occurs more frequently during specific time windows, suggesting behavioural exploitation.
- **Feature relationships:** Correlation analysis showed low multicollinearity among numerical features, supporting model stability.

EDA confirmed that meaningful patterns exist that machine learning models can exploit.

5. Preprocessing and Feature Engineering

Categorical variables were encoded to make them compatible with machine learning algorithms. The dataset was split into training and testing sets using stratification to preserve the fraud ratio.

To address class imbalance, **SMOTENC** was applied. This technique is particularly suitable because it generates synthetic samples while respecting both numerical and categorical features. The result was a balanced training dataset that improved the model's ability to learn fraud patterns without leaking information into the test set.

Numerical features were scaled to ensure fair contribution to model training and to improve optimisation stability.

6. Model Training

XGBoost was selected as the primary algorithm due to its strong performance on tabular data, ability to model non-linear relationships, and robustness to noise. Hyperparameter tuning was performed using GridSearchCV with cross-validation to optimise model performance and generalisability.

Key tuned parameters included learning rate, tree depth, and the number of estimators. This systematic approach ensured that the final model was not overfit to the training data.

7. Model Evaluation

Model performance was evaluated using metrics appropriate for imbalanced classification:

- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion matrix analysis

The model demonstrates strong recall, which is critical in fraud detection due to the high business cost associated with missed fraudulent transactions. The F1-score closely aligns with recall, indicating that the model prioritises fraud capture while maintaining a reasonable balance between precision and recall under severe class imbalance. Precision remains at an acceptable level, reflecting a practical trade-off between effective fraud prevention and minimising unnecessary customer friction. Although the ROC-AUC score of 0.6311 indicates moderate rather than exceptional discriminative capability, this performance is considered reasonable for a highly imbalanced fraud detection task using synthetic data and remains aligned with the primary business objective of minimising undetected fraud.

8. Explainability with SHAP

To ensure transparency, SHAP (Shapley Additive Explanations) was used to interpret model predictions. SHAP analysis revealed that transaction amount and engineered time-based features—such as hour of day and weekend indicators—were among the most influential predictors, aligning with patterns observed during exploratory data analysis and domain expectations.

The model provides transparent, **local (transaction-level) explanations** for its fraud predictions, allowing institutions to clearly understand and justify individual decisions. This level of interpretability supports auditing and regulatory compliance and helps build trust in the fraud detection system.

9. Conclusion

This project demonstrates that machine learning techniques can be effectively applied to detect fraudulent transactions within a transaction environment inspired by the South African financial ecosystem. Through comprehensive exploratory data analysis, appropriate handling of class imbalance, and the use of a robust tree-based model, the system can capture meaningful and realistic fraud-related patterns. Despite relying on a synthetic dataset, the model achieves a strong recall, which is critical in fraud detection scenarios where the cost of missed fraudulent activity is high. Additionally, the incorporation of explainability techniques enables transparent, transaction-level insights

into model predictions, thereby supporting trust, auditability, and regulatory considerations.

With access to real-world transaction data and the implementation of continuous monitoring and retraining strategies, this approach has the potential to evolve into a production-grade fraud detection system. The methodology demonstrated in this project provides a solid foundation for deployment in real operational settings and aligns well with industry best practices. Overall, the project successfully meets its stated objectives and represents a strong, practical example of applied data science, making it suitable for both real-world applications.

10. Recommendations

- The model should be used as a decision-support tool that allows organisations to benefit from fraud detection insights while maintaining human oversight and customer experience.
- Continuously retrain the model to account for evolving fraud strategies.
- Incorporate additional behavioural and device-level features where legally permissible.
- Combine machine learning predictions with rule-based systems for layered defence.
- Implement cost-sensitive threshold tuning to align with business priorities.